

Relations

Section 8.1, 8.3—8.5 of Rosen

CSCE 235 Introduction to Discrete Structures

Course web-page: cse.unl.edu/~cse235

Relations, Digraphs (07)

- 3.1 Relations, Paths and Digraphs
- 3.2 Properties and types of binary relations
- 3.3 Manipulation of relations, Closures, Warshall's algorithm
- 3.4 Equivalence relations

Introduction

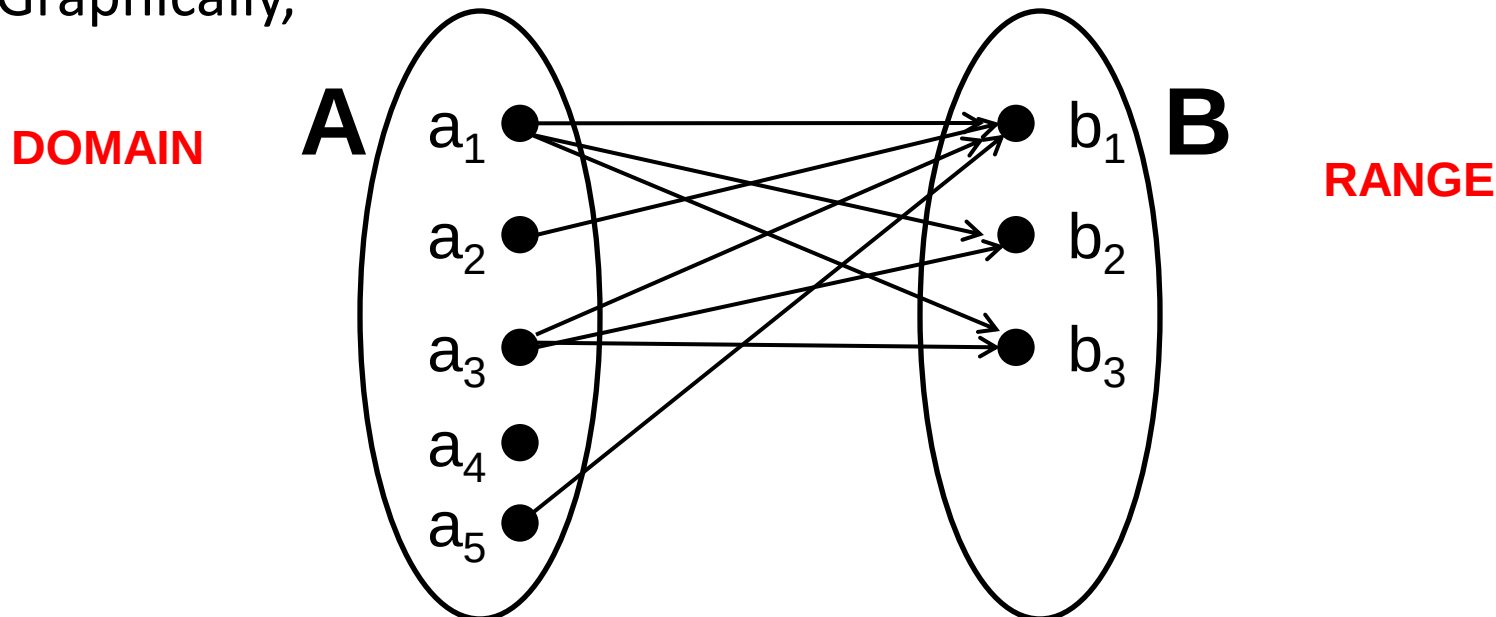
- A relation between elements of two sets is a subset of their Cartesian products (set of all ordered pairs)
- **Definition:** A binary relation from a set A to a set B is a subset
$$R \subseteq A \times B = \{ (a,b) \mid a \in A, b \in B \}$$
- When $(a,b) \in R$, we say that a is related to b.
- Notation: aRb , ~~aRb~~

Relations: Representation

- To represent a relation, we can enumerate every element of R
- Example
 - Let $A=\{a_1,a_2,a_3,a_4,a_5\}$ and $B=\{b_1,b_2,b_3\}$
 - Let R be a relation from A to B defined as follows

$$R=\{(a_1,b_1),(a_1,b_2),(a_1,b_3),(a_2,b_1),(a_3,b_1),(a_3,b_2),(a_3,b_3),(a_5,b_1)\}$$

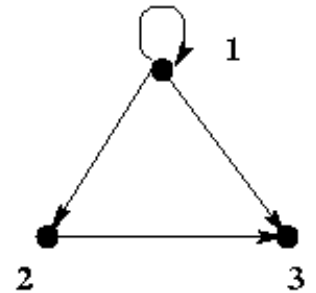
Graphically,



DIGRAPHS-Directed Graphs

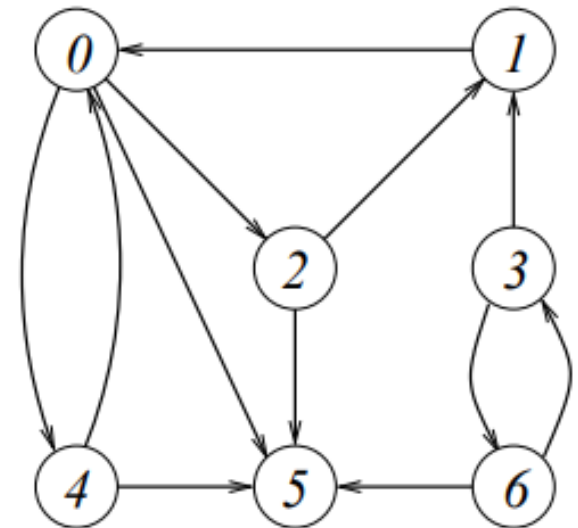
A **digraph** (directed graph) is a diagram composed of points called **vertices** (nodes) and arrows called **edges** going from a vertex to a vertex.

Example :-A digraph with 3 vertices and 4 edges



Example: - $V = \{0, 1, 2, 3, 4, 5, 6\}$, $E = \{(0, 2), (0, 4), (0, 5), (1, 0), (2, 1), (2, 5), (3, 1), (3, 6), (4, 0), (4, 5), (6, 3), (6, 5)\}$

Matrix Representation ?

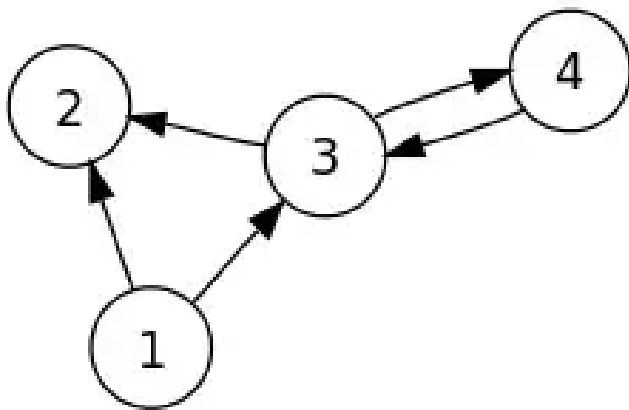


Degree of Vertex in a Directed Graph

A directed graph, each vertex has an **in-degree** and an **out-degree**.

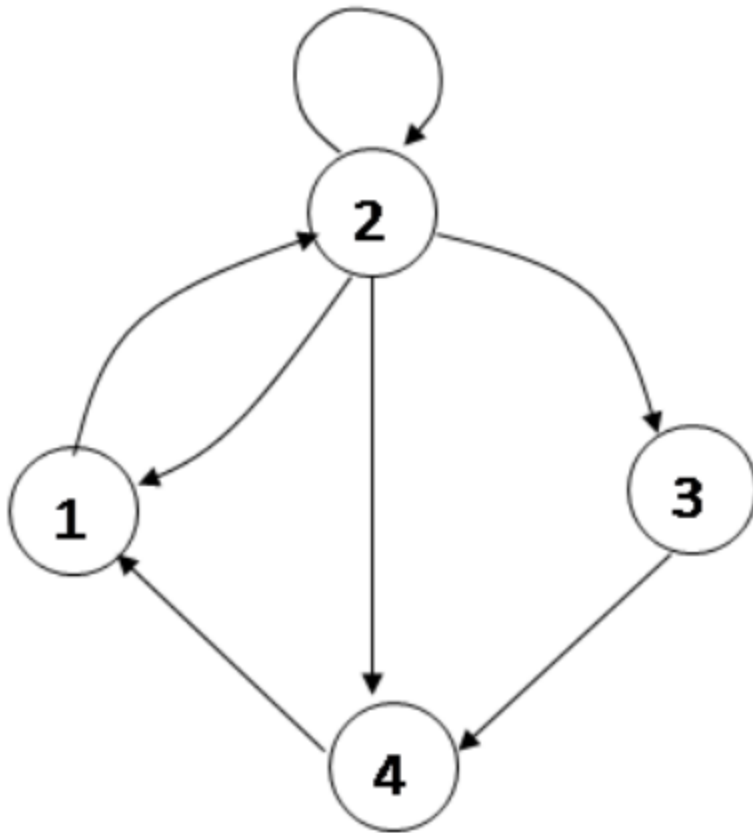
In-degree of a Graph-Number of edges which are coming into the vertex V.

Out-degree of a Graph-Number of edges which are going out from the vertex V



VERTEX	1	2	3	4
In Degree	0	2	2	1
Out-degree	2	0	2	1

Find out in degree and out degree

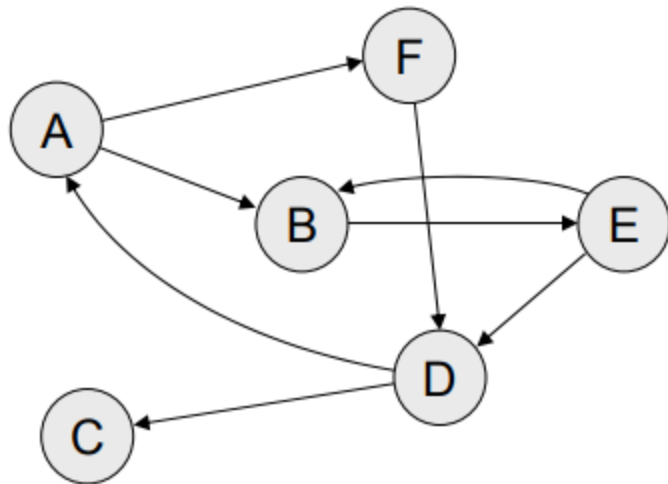


VERTEX	1	2	3	4
In Degree	2	2	1	2
Out-degree	1	4	1	1

Problems

For the digraph shown let R be given by digraph shown.

Find M_R and R

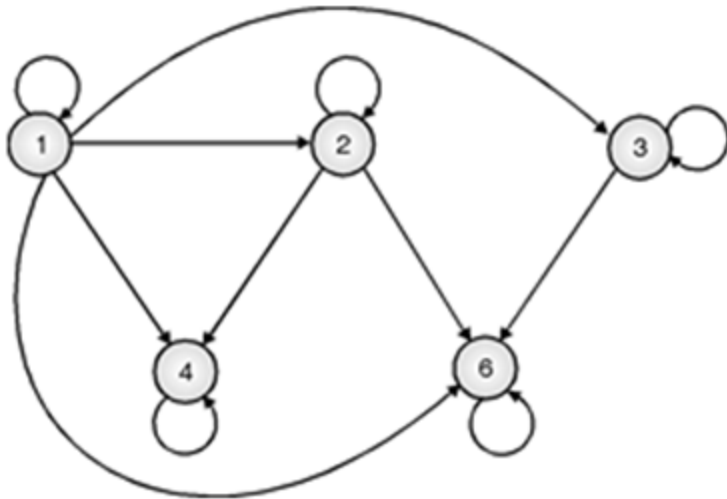


	A	B	C	D	E	F
A	0	1	0	0	0	1
B	0	0	0	0	1	0
C	0	0	0	0	0	0
D	1	0	1	0	0	0
E	0	1	0	1	0	0
F	0	0	0	1	0	0

Example

Let $A = \{1, 2, 3, 4, 6\}$ and let R be the relation on A defined by ' x divides y '. Find R and draw the digraph of R . Find Matrix of R .

Soln.: $R = \{(1,1), (1,2), (1,3), (1,4), (1,6), (2,2), (2,4), (2,6), (3,3), (3,6), (4,4), (6,6)\}$



Assume the rows and columns of M are each labelled 1, 2, 3, 4, 6, since R is relation from A to A , the matrix M_R is square, i.e. M_R has the same number of row as column

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 6 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 6 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

Example

Let $A = \{1, 2, 3, 4, 6\} = B$, $a R b$ if and only if a is a multiple of b . Find R and draw the digraph of R . Find Matrix of R .

Solution:

$$R = \{(1, 1), (2, 1), (2, 2), (3, 1), (3, 3), (4, 1), (4, 2), (4, 4), (6, 1), (6, 2), (6, 3), (6, 6)\}$$

$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 6 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 6 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix} \end{matrix}$$

Problems

1. Draw the graphical representation of relation 'less than' on $\{1, 2, 3, 4\}$

$$R = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$$

2. $A = \{2, 3, 4, 5\}$,

$$R = \{(2, 3), (3, 2), (3, 4), (3, 5), (4, 3), (4, 4), (4, 5)\}$$

Draw Digraph

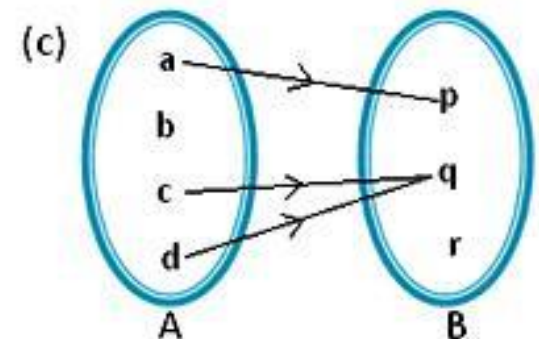
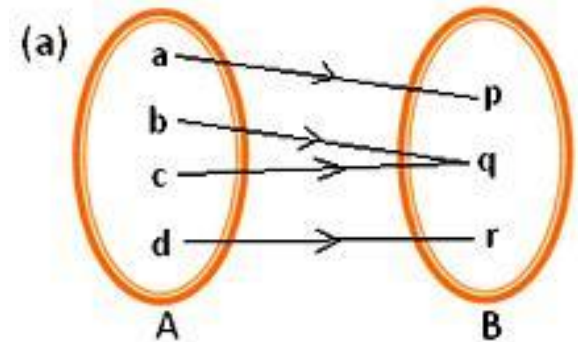
→ Domain, Range of Relation R

$$\text{Ex : } A = \{a, b, c, d\}, B = \{1, 2, 3\}$$

$$R = \{(a, 1), (a, 2), (b, 1), (c, 2), (d, 1)\}$$

$$\text{Dom}(R) = \{a, b, c, d\}$$

$$\text{Ran}(R) = \{1, 2\}$$



Problems

1. Let $A = \{ 1, 2, 3, 4, 8 \} = B$ only **if $a=b$** .

Find the relation R , draw digraph and also write M_R

2. Let $A = \{ 1, 2, 3, 4, 8 \} = B$

$a R b$ iff a is **a multiple of b**

$a R b$ iff **$a + b \leq 9$**

Find the relation R , draw digraph and also write M_R

3. Let $A = \{ 1, 3, 5, 7, 9 \}$, $B = \{ 2, 4, 6, 8 \}$; $a R b$ iff **$b < a$**

PATHS

$R = \{ (1, 2), (2, 3), (2, 4), (3, 3) \}$ is a relation on $A = \{1, 2, 3, 4\}$

$$R^1 = R = \{ (1, 2), (2, 3), (2, 4), (3, 3) \}$$

$$R^2 = \{ (1, 3), (1, 4), (2, 3), (3, 3) \}$$

$1 R^2 3$ Since $1 R 2$ and $2 R 3$

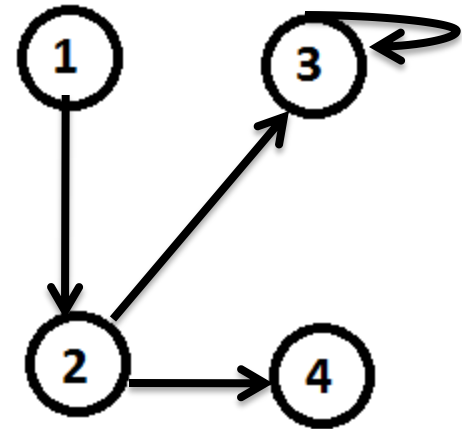
$1 R^2 4$ Since $1 R 2$ and $2 R 4$...

$$R^3 = \{ (1, 3), (2, 3), (3, 3) \}$$

$$R^4 = \{ (1, 3), (2, 3), (3, 3) \}$$

R infinity is all ordered pairs where there is a path of any length

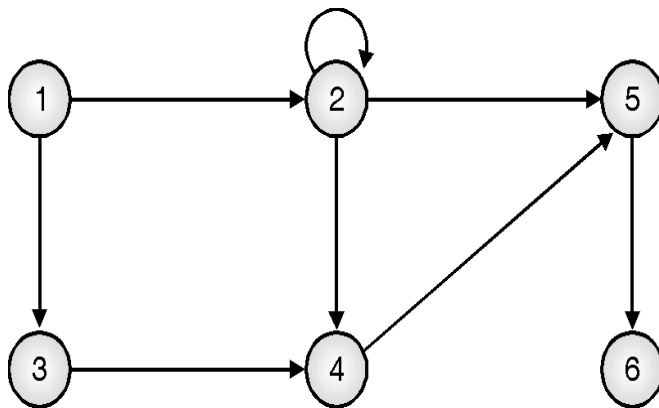
$$R^\infty = \{ (1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 3) \}$$



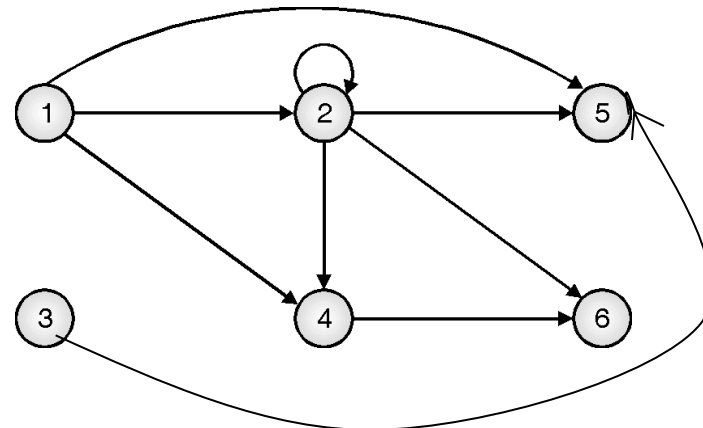
Paths in Relations and Digraphs

Let $A = \{1, 2, 3, 4, 5, 6\}$. Let R be the relation whose digraph is shown in Fig.

Find R^2 and draw digraph of the relation R^2 .



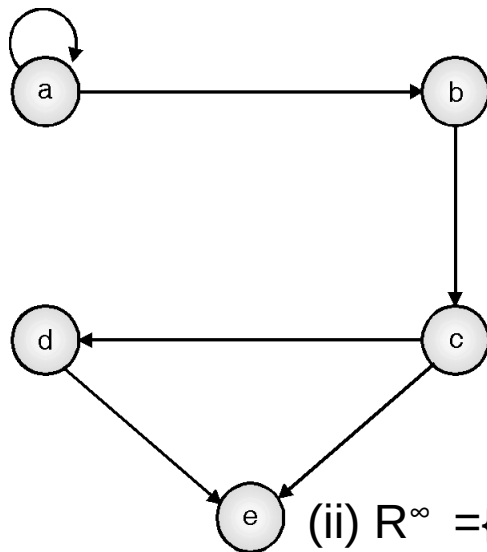
$1 R^2 2$	Since	$1 R 2$	and	$2 R 2$
$1 R^2 4$	Since	$1 R 2$	and	$2 R 4$
$1 R^2 5$	Since	$1 R 2$	and	$2 R 5$
$2 R^2 2$	Since	$2 R 2$	and	$2 R 2$
$2 R^2 4$	Since	$2 R 2$	and	$2 R 4$
$2 R^2 5$	Since	$2 R 2$	and	$2 R 5$
$2 R^2 6$	Since	$2 R 5$	and	$5 R 6$
$3 R^2 5$	Since	$3 R 4$	and	$4 R 5$
$4 R^2 6$	Since	$4 R 5$	and	$5 R 6$



Paths in Relations and Digraphs

Let $A = \{a, b, c, d, e\}$
and $R = \{(a, a), (a, b), (b, c), (c, e), (c, d), (d, e)\}$

Compute (i) R^2 (ii) R^∞



$a R^2 a$	Since	$a R a$	and	$a R a$
$a R^2 b$	Since	$a R a$	and	$a R b$
$a R^2 c$	Since	$a R b$	and	$b R c$
$b R^2 e$	Since	$b R c$	and	$c R e$
$b R^2 d$	Since	$b R c$	and	$c R d$
$c R^2 e$	Since	$c R d$	and	$d R e$

(ii) $R^\infty = \{(a, a), (a, b), (a, c), (a, d), (a, e), (b, c), (b, d), (b, e), (c, d), (c, e), (d, e)\}$

PROBLEMS

1. Let $A = \{1, 2, 3, 4, 5\}$ and R be relation defined by $a R b$ iff $a < b$ compute R, R^2, R^3 Draw digraph of R, R^2 and R^3

$$R = \{(1, 2), (1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)\}$$

$$R^2 = \{(1, 3), (1, 4), (1, 5), (2, 4), (2, 5), (3, 5)\}$$

$$R^3 = \{(1, 4), (1, 5), (2, 5)\}$$

2. Consider $R = \{(1, 1), (2, 1), (3, 2), (4, 3)\}$

Compute R^2, R^3, R^4

3. Let $A = \{a, b, c, d, e\}$, $R = \{(a, a), (a, b), (b, c), (c, e), (c, d), (d, e)\}$

Draw digraph of R, M_R , Compute R^∞

Properties/Types of Relations

- Reflexive
- Symmetric
- Transitive
- Antisymmetric
- Asymmetric

Properties: Reflexivity

- In a relation on a set, if all ordered pairs (a,a) for every $a \in A$ appears in the relation, R is called reflexive
- **Definition:** A relation R on a set A is called reflexive iff

$$\forall a \in A (a, a) \in R$$

– Eg: $A = \{1, 2, 3\}$,

$$R = \{(1, 1), (2, 2), (3, 3)\}$$

– Irreflexive ?

Assume the relation R on $A = \{1, 2, 3, 4\}$ Is R_1/R_2 irreflexive?

$$R_1 = \{(1, 2), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (3, 4), (4, 1), (4, 2), (4, 3)\}$$

$$R_2 = \{(1, 2), (2, 2), (3, 3)\}$$

Properties: Symmetry

- **Definitions:**

- A relation R on a set A is called symmetric if whenever **$a R b$ and $b R a$** i.e

$$\forall a, b \in A \quad ((b, a) \in R \Leftrightarrow (a, b) \in R)$$

Eg 1 : $A = \{ 1, 2, 3 \}$, Is R symmetric ?

$$R = \{ (1, 2), (2, 1), (2, 3), (3, 2), (1, 1) \}$$

Eg 2 : $A = \{ 1, 2, 3, 4 \}$, Is R symmetric ?

$$R = \{ (1, 2), (1, 3), (1, 4), (2, 1), (2, 3), (2, 4), (3, 1), (3, 2), (3, 4), (4, 1), (4, 2), (4, 3) \}$$

Asymmetric relation: Asymmetric relation is opposite of symmetric relation.

A relation R on a set A is called asymmetric if no $(b,a) \in R$ when $(a,b) \in R$

AntiSymmetric Relation: A relation R on a set A is called antisymmetric if $(a,b) \in R$ and $(b,a) \in R$ **if** $a = b$ is called antisymmetric.i.e.

UNLESS there exists $(a, b) \in R$ and $(b, a) \in R$, AND $a \neq b$

Eg : $A = \{ 1, 2, 3, 4 \}$ and $R = \{ (1, 2), (2, 2), (3, 3) \}$

Is R anti-symmetric?

Answer: Yes. It is anti-symmetric as 2,1 is not there

Symmetry versus Antisymmetry

- In a symmetric relation $aRb \Leftrightarrow bRa$
- In an antisymmetric relation, if we have aRb and bRa hold only when $a=b$
- An antisymmetric relation is not necessarily a reflexive relation
- **A relation that is not symmetric is not necessarily asymmetric**
- **An anti-symmetric relation is a binary relation where the following two conditions are met:**
 - **1) If A is related to B, then B cannot be related to A.**
 - **2) If A is not related to B, then B cannot be related to A.**
- In Maths, we can conclude that a binary relation on a set is called as antisymmetric if there is no pair of distinct elements.

Properties: Transitivity

- **Definition:** A relation R on a set A is called transitive if whenever $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R$ for all $a,b,c \in A$

$$\forall a, b, c \in A ((a R b) \wedge (b R c)) \Rightarrow a R c$$

Example

$R = \{ (1, 2), (2, 3), (1, 3) \}$ on set

$A = \{ 1, 2, 3 \}$ is transitive.

Special cases

1) Let $A = \{ 1, 2, 3, 4 \}$

$$R = \{ (1, 2), (1, 3), (4, 2) \}$$

Is R transitive?

YES

2) $R = \{ \}$

3) A relation that is symmetric and anti-symmetric

$$R = \{(1,1), (2,2)\} \text{ on the set } A = \{1,2,3\}$$

Properties of Relations

State whether R satisfies property of reflexive , irreflexive , symmetry, asymmetry , antisymmetry , transitivity for $A=\{1,2,3,4\}$

1. $R=\{(1,1),(1,2),(2,1),(2,2),(3,3),(4,3),(3,4),(4,4)\}$ **R,S,T,**
2. $R= \{(1,3),(1,1),(3,1),(1,2),(3,3),(4,4)\}$
3. $R=\{(1,1),(1,2),(2,1),(2,2),(3,3),(3,4)\}$
4. $R=\{(1,3),(1,4),(2,3),(2,4),(3,1),(3,4)\}$
5. $R=\{(1,1),(2,2),(3,3),(4,4)\}$

EQUIVALENCE RELATION

A relation is an **Equivalence Relation** if it is **REFLEXIVE, SYMMETRIC, AND TRANSITIVE.**

Let $A = \{ a , b , c \}$ and

$R = \{ (a , a), (b , b), (b , c), (c , b), (c , c) \}$

is an equivalence relation since it is

REFLEXIVE, SYMMETRIC, & TRANSITIVE.

Determine whether R is an Equivalence relation

$$1) R = \{ (1, 1), (2, 2), (3, 3), (1, 2), (2, 1), (2, 3), (3, 2), (1, 3), (3, 1) \}$$

on set $A = \{ 1, 2, 3 \}$

$$2) A = \{ 1, 2, 3, 4 \}, R = \{ (1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 3), (3, 3), (4, 4) \}$$

$$3) \text{ Let } A = \{ a, b, c, d \}$$

$$R = \{ (a, a), (b, a), (b, b), (c, c), (d, d), (d, c) \}$$

Determine whether R is an Equivalence relation

Let $A = \{a, b, c\}$ and let , $M_R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

Determine whether R is an equivalence relation.

Soln.: $R = \{(a, a), (b, b), (b, c), (c, b), (c, c)\}$

R is reflexive since $(a, a), (b, b), (c, c) \in R$

R is symmetric since $(b, c) \in R \rightarrow (c, b) \in R$

R is transitive since,

(b, b)	and	$(b, c) \in R$	implies	$(b, c) \in R,$
(b, c)	and	$(c, b) \in R$	implies	$(b, b) \in R,$
(c, c)	and	$(c, b) \in R$	implies	$(c, b) \in R,$
(c, b)	and	$(b, b) \in R$	implies	$(c, b) \in R,$
(c, b)	and	$(b, c) \in R$	implies	$(c, c) \in R,$
(b, c)	and	$(c, c) \in R$	implies	$(b, c) \in R,$

Hence R is an equivalence relation.

Equivalence Class and Partitions

- Let $A = \{1, 2, 3, 4\}$ and consider the partition

$$P = \{\{1, 2, 3\}, \{4\}\} \text{ of } A.$$

Find the equivalence relation R on A determined by P

“ Each element in a block is related to every other element in the same block and only to those elements “

$$R = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3), (4,4)\}$$

Problems

Find the equivalence relation on A by P and construct its **digraph**

1) Let $A = \{a, b, c, d\}$ and $P = \{\{a, b\}, \{c\}, \{d\}\}$

2) Let $A = \{1, 2, 3, 4, 5\}$ and $P = \{\{1, 2\}, \{3\}, \{4, 5\}\}$

$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4), (4, 5), (5, 4), (5, 5)\}$

3) If $\{\{1, 3, 5\}, \{2, 4\}\}$ is a partition on the set $A = \{1, 2, 3, 4, 5\}$, determine the corresponding equivalence relation

$R = \{(1, 1), (3, 3), (5, 5), (1, 3), (1, 5), (3, 5), (3, 1), (5, 1), (5, 3), (2, 2), (4, 4), (2, 4), (4, 2)\}$

EQUIVALENCE CLASS

Let $A = \{1, 2, 3, 4, 5, 6\}$ and let R be the equivalence relation on A defined by

$$R = \{(1, 1), (1, 5), (2, 2), (2, 3), (2, 6), (3, 2), (3, 3), (3, 6), (4, 4), (5, 1), (5, 5), (6, 2), (6, 3), (6, 6)\}$$

Find the equivalence classes of R and find the partition of A induced by R

$$R = \{(1,1), (1,5), (2,2), (2,3), (2,6), (3,2), (3,3), (3,6), (4,4), (5,1), (5,5), (6,2), (6,3), (6,6)\}$$

Equivalence Classes:

$$R(1) = \{1, 5\}$$

$$R(2) = \{2, 3, 6\}$$

$$R(3) = \{2, 3, 6\}$$

$$R(4) = \{4\}$$

$$R(5) = \{1, 5\}$$

$$R(6) = \{2, 3, 6\}$$

Therefore, the partition of A induced by R i.e

$$A/R = \{\{1, 5\}, \{2, 3, 6\}, \{4\}\}$$

Rank R (Number of distinct equivalence classes)
= 3

Problems

1. Let $A=\{1,2,3\}$ and let $R=\{(1,1),(2,2),(1,3),(3,1),(3,3)\}$.

Find $A|R$.

2. Let $A = \{1,2,3,4\}$, and let

$R=\{(1,1),(1,2),(2,1),(2,2),(3,4),(4,3),(3,3),(4,4)\}$

Determine $A|R$.

3. Let $A = \{1,2,3,4\}$, and let

$R=\{(1,1),(1,2),(1,3),(2,1),(2,2),(3,1),(2,3),(3,2),(3,3),(4,4)\}$

Show that R is an equivalence relation and determine the equivalence classes and hence find $A|R$ and rank of R

Combining Relations

- Relations are simply... sets (of ordered pairs); subsets of the Cartesian product of two sets
- Therefore, in order to combine relations to create new relations, it makes sense to use the usual set operations
 - **Compliment R**
 - **Intersection $(R_1 \cap R_2)$**
 - **Union $(R_1 \cup R_2)$**
 - **Set difference $(R_1 \setminus R_2)$**
 - **Inverse R^{-1}**

Example: Let $A = \{1, 2, 3\}$ and $B = \{u, v\}$ and

$$R1 = \{(1, u), (2, u), (2, v), (3, u)\}$$

and

$$R2 = \{(1, v), (3, u), (3, v)\}$$

$$R1 \cup R2 =$$

$$\{(1, u), (1, v), (2, u), (2, v), (3, u), (3, v)\}$$

$$R1 \cap R2 =$$

$$\{(3, u)\}$$

$$R1 - R2 =$$

$$\{(1, u), (2, u), (2, v)\}$$

$$R2 - R1 =$$

$$\{(1, v), (3, v)\}$$

$A = \{a, b, c, d\}$ and

$R = \{(a, b), (b, c), (a, c), (c, d)\}$ then

$R^{-1} = \{(b, a), (c, b), (c, a), (d, c)\}$

Let $A = \{ 1, 2, 3, 4 \}$ and $B = \{ a, b, c \}$ and let

$R = \{(1,a), (1,b), (2,b), (2,c), (3,b), (4,a)\}$ and

$S = \{(1,b), (2,c), (3,b), (4,b)\}$

Compute $R \cap S$, $R \cup S$, S^{-1} and R^{-1}

Combining Relations: Example

- Let
 - $A = \{1, 2, 3, 4\}$
 - $B = \{1, 2, 3, 4\}$
 - $R_1 = \{(1, 2), (1, 3), (1, 4), (2, 2), (3, 4), (4, 1), (4, 2)\}$
 - $R_2 = \{(1, 1), (1, 2), (1, 3), (2, 3)\}$
- Let
 - $R_1 \cup R_2 =$
 - $R_1 \cap R_2 =$
 - $R_1 - R_2 =$
 - $R_2 - R_1 =$

Composite of Relations

- **Definition:** Let R_1 be a relation from the set A to B and R_2 be a relation from B to C , i.e.

$$R_1 \subseteq A \times B \text{ and } R_2 \subseteq B \times C$$

the composite of R_1 and R_2 is the relation consisting of ordered pairs (a,c) where $a \in A$, $c \in C$ and for which there exists an element $b \in B$ such that $(a,b) \in R_1$ and $(b,c) \in R_2$. We denote the composite of R_1 and R_2 by

$$R_2 \circ R_1$$

Ex: Let $A = \{ 1, 2, 3 \}$, $B = \{ 0, 1, 2 \}$ and $C = \{ a, b \}$

$R = \{ (1, 0), (1, 2), (3, 1), (3, 2) \}$

$S = \{ (0, b), (1, a), (2, b) \}$

$S \circ R = ?$

$\{ (1, b), (3, a), (3, b) \}$

Since $(1, 0) \in R$ and $(0, b) \in S$, $\therefore (1, b) \in S \circ R$

Since $(1, 2) \in R$ and $(2, b) \in S$, $\therefore (1, b) \in S \circ R$

Since $(3, 1) \in R$ and $(1, a) \in S$, $\therefore (3, a) \in S \circ R$

Since $(3, 2) \in R$ and $(2, b) \in S$, $\therefore (3, b) \in S \circ R$

Problems

1. Let $A=\{1,2,3\}$ and let

$R=\{(1,1),(1,3),(2,1),(2,2),(2,3),(3,2)\}$ and

$S=\{(1,1),(2,2),(2,3),(3,1),(3,3)\}$.

Find M_{SoR}

$\text{SoR}=\{(1,1),(1,3),(2,1),(2,2),(2,3),(3,2),(3,3)\}$

2. Let $A=\{1,2,3,4\}$

$R=\{(1,1),(1,2),(2,3),(2,4),(3,4),(4,1),(4,2)\}$

$S=\{(3,1),(4,4),(2,3),(2,4),(1,1),(1,4)\}$

Compute $\text{SoR}, \text{RoS}, \text{RoR}, \text{SoS}$

$\text{SoR}=\{(1,1),(1,3),(2,1),(2,4),(3,4),(4,1),(4,4),(1,4),(4,3)\}$

$\text{RoS}=\{(3,1),(3,2),(4,1),(4,2),(2,4),(2,1),(2,2),(1,1),(1,2)\}$

$\text{RoR}=13$ elements

$\text{SoS}=7$ elements

Reflexive closure

Let R be a relation on a set A , and R is not reflexive (i.e. some pairs of the diagonal relation Δ are not in R).

A relation $R_1 = R \cup \Delta$ is the reflexive closure of the relation R if $R \cup \Delta$ is the smallest relation containing R which is reflexive.

$R_1 = R \cup \Delta$ where Δ is the set of elements of the type (a, a)

where $a \in A$.

Example -Reflexive closure

A = {1, 2, 3} and the relation R is given by

R = {(1, 1), (1, 2), (2, 3)} then

$R_1 = R \cup \Delta$ where

$\Delta = \{(1, 1), (2, 2), (3, 3)\}$

$R \cup \Delta = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 3)\}$

Reflexive closure is,

$R_1 = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 3)\}$

Symmetric closure

Suppose that R is a relation on A that is not symmetric.

Then there must exist pairs (x, y) in R such that (y, x) is not in R . Of course, $(y, x) \in R^{-1}$, so if R is to be symmetric we must add all pairs from R^{-1} ;

that is we must enlarge R to $\mathbf{R \cup R^{-1}}$.

so $R \cup R^{-1}$ is the smallest symmetric relation containing R ;

that is $R \cup R^{-1}$ is the 'symmetric closure' of R .

example

$A = \{a, b, c, d\}$ and

$R = \{(a, b), (b, c), (a, c), (c, d)\}$ then

$R^{-1} = \{(b, a), (c, b), (c, a), (d, c)\}$

so the symmetric closure of R is

$R \cup R^{-1} = \{(a, b), (b, a), (b, c), (c, b), (a, c), (c, a), (c, d), (d, c)\}$

Transitive closure

Let R be a relation on a set A . Then the 'transitive closure' of a relation R is the smallest transitive relation containing R . The transitive closure of R is just the connectivity relation R^∞ .

$R^* = \text{Transitive closure of } R$

$$= R \cup \{(a, c), \text{ if and only if } (a, b), (b, c) \in R\}$$

example

Find the transitive closure R^* of the relation R on $A = \{1, 2, 3, 4\}$ defined by the directed graph shown

Soln.:

$R = \{(1, 3), (1, 4), (3, 2), (3, 3), (3, 4)\}$

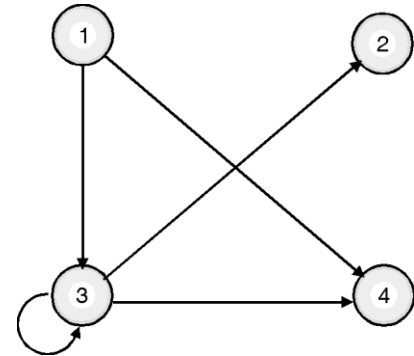
Here transitive closure of R is

$= R \cup \{(a, c) \mid \text{if } (a, b), (b, c) \in R\}$

To find transitive closure

$(1, 3) \in R$ and $(3, 2) \in R$, hence add $(1, 2)$ in R

Transitive closure of $R = \{(1, 2), (1, 3), (1, 4), (3, 2), (3, 3), (3, 4)\}$



Warshall's algorithm

Ex. 1 : Let $A = \{1, 2, 3, 4\}$ and let $R = \{(1, 2), (2, 3), (3, 4), (2, 1)\}$.

Find transitive closure of R using Warshall's algorithm.

Solution:

$$W_0 = M_R = \begin{bmatrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{bmatrix}$$

First we find W_1 , so that $k = 1$. W_0 has 1's in location 2 of column 1 i.e. $(2, 1)$ and location 2 of row 1 i.e. $(1, 2)$

$i \quad j$
 $p_1: (2, 1)$
 $i \quad j$
 $q_1: (1, 2)$
 add (p_i, q_j) i.e. $(2, 2)$ in W_k

Thus W_1 is just W_0 with a new 1 in position $(2, 2)$

$$W_1 = \begin{bmatrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{bmatrix}$$

$$W_1 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

Matrix W_1 has 1's at row 1 and 2 of column 2 and columns 1, 2, and 3 of row 2. i.e.

$$\begin{matrix} i & j \\ p_1 & : & (1, 2) & p_2 & : & (2, 2) \\ i & j & i & j & i & 1 \\ q_1 & : & (2, 1) & q_2 & : & (2, 2) & q_3 & : & (2, 3) \end{matrix}$$

i.e. (1, 1), (1, 2), (1, 3), (2, 1), (2, 2) and (2, 3) of matrix W_1 (if 1's are not already there).

$$W_2 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$p_1 : (1, 3) \quad p_2 : (2, 3)$$

$$i \quad j$$

$$q_1 : (3, 4)$$

$$W_3 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Finally, W_3 has 1's in locations 1, 2, 3 of column 4 and no 1's in row 4, so no new 1's are added and $MR_\infty = W_4 = W_3$.

$$M_R = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$M_S = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\text{So } M_{R \cup S} = M_R \vee M_S = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

We now compute $M_{(R \cup S)^\infty}$ by Warshall's algorithm. First, $W_0 = M_{R \cup S}$. We next compute W_1 so $k = 1$. Since W_0 has 1's in locations 1 and 2 of column 1 and in locations 1 and 2 of row 1, we find that no new 1's must be adjoined to W_1 . Thus

$$W_0 = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

K=1

$$\begin{array}{cc} & \begin{matrix} i & j \end{matrix} \\ p_1 : & (1, 1) \end{array} \quad \begin{array}{cc} & \begin{matrix} i & j \end{matrix} \\ p_2 : & (2, 1) \end{array}$$

$$\begin{array}{cc} & \begin{matrix} i & j \end{matrix} \\ q_1 : & (1, 1) \end{array} \quad \begin{array}{cc} & \begin{matrix} i & j \end{matrix} \\ q_2 : & (1, 2) \end{array}$$

To obtain W_1 , we must put 1's in positions (1, 1), (1, 2), (2, 1) and (2, 2). We see that

$$W_1 = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Thus $W_1 = W_0$

We now compute W_2 , so $k = 2$.

Since W_1 has 1's in locations 1 and 2 : of column 2 and in locations 1 and 2 of row 2, we find that no new 1's must be added to W_1 . That is,

$$\begin{array}{cc} \begin{array}{c} i \quad j \\ p_1 : (1, 2) \end{array} & \begin{array}{c} i \quad j \\ p_2 : (2, 2) \end{array} \\ \\ \begin{array}{c} i \quad j \\ q_1 : (2, 1) \end{array} & \begin{array}{c} i \quad j \\ q_2 : (2, 2) \end{array} \end{array}$$

To obtain W_2 , we must put 1's in positions (1, 1), (1, 2), (2, 1), (2, 2). We see that

$$W_2 = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Thus $W_2 = W_1$

We next compute W_3 , so $k = 3$. Since W_2 has 1's in locations 3 and 4 of column 3 and in locations 3 and 4 of row 3, we find that no new 1's must be added to W_2 . That is

$$\begin{array}{cc} \begin{array}{c} i \quad j \\ p_1 : (3, 3) \end{array} & \begin{array}{c} i \quad j \\ p_2 : (4, 3) \end{array} \\ \\ \begin{array}{c} i \quad j \\ q_1 : (3, 3) \end{array} & \begin{array}{c} i \quad j \\ q_2 : (3, 4) \end{array} \end{array}$$

To obtain W_3 , we must put 1's in position (3, 3), (3, 4), (4, 3), (4, 4). We see that

$$W_3 = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Thus $W_3 = W_2$

Things change when we now compute W_4 . Since W_3 has 1's in locations 3, 4, and 5 of column 4 and in locations 3, 4 and 5 of column 4, and in locations 3, 4 and 5 of row 4 we must add new 1's to W_3 in positions 3, 5, and 5, 3, i.e.

$$\begin{array}{ccc} \begin{array}{c} i \quad j \\ p_1 : (3, 4) \end{array} & \begin{array}{c} i \quad j \\ p_2 : (4, 4) \end{array} & \begin{array}{c} i \quad j \\ p_3 : (5, 4) \end{array} \\ \begin{array}{c} i \quad j \\ q_1 : (4, 3) \end{array} & \begin{array}{c} i \quad j \\ q_2 : (4, 4) \end{array} & \begin{array}{c} i \quad j \\ q_3 : (4, 5) \end{array} \end{array}$$

To obtain W_4 , we must put 1's in positions (3, 3), (3, 4), (3, 5), (4, 3), (4, 4), (4, 5), (5, 3), (5, 4), (5, 5). We see that,

$$W_4 = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

You may verify that $W_5 = W_4$ and thus

$$(R \cup S)^\infty = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3, 4), (3, 5), (4, 3), (4, 4), (4, 5), (5, 3), (5, 4), (5, 5)\}$$

Transitive Closure and Warshall's Algorithm

Compute the Warshall's Algorithm transitive closure of

- $R = \{(a,b), (b,c), (c,d), (b,a)\}$ on set $A = \{a,b,c,d\}$
- $R = \{(1,1), (1,2), (1,4), (2,2), (2,3), (3,1), (3,4), (4,1), (4,4)\}$ on the set $A = \{1,2,3,4\}$

Computer transitive closure using Warshall's algorithm where $A=\{a_1, a_2, a_3, a_4, a_5\}$ and R be a relation on A whose matrix is

$$M_R = W_0 = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$